

§ 4044 Asset Allocation

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Presented and Edited by Tim Carlson and Dwayne Jeffers,

Background for Asset Allocation

- Qualified Pre-Retirement Survivor Annuity
- Expense Load
- Valuation Assets

Valuing the Qualified Pre-Retirement Survivor Annuity (QPSA)

For married non-Retirees (including Non-Vested) with deferred XRAs:

- If Participant bears full cost of QPSA at DOPT, do not value QPSA.
- If no QPSA charge at DOPT, value QPSA in each PC that has Participant benefit liability.

Formula: $(12) \times (\text{Survivor's monthly benefit in PC}) \times (\text{Probability Participant dies between ANB and XRA}) \times (\text{Present value factor for Spouse's Straight Life Annuity deferred from DOPT to XRA with no Spouse mortality in deferral period})$ - See [ATM III.G.QPSA](#) for more details

PBGC Expense Load PVPBL

- Expense Load is defined in [4044 Appendix C](#)
- Expressed as a percentage of Plan Liabilities
- Each Priority Category with a Benefit Liability has an expense load applied

PBGC Expense Load if PVPBL $\leq$ \$200,000

Expense Loading (4044 Appendix C):

The expense load is calculated as follows:

1. 5% of PVPBL without load
2. \$200 x (Number of valuation participants)
3. Loading percentage = $[(1) + (2)] / \text{PVPBL without load}$

PBGC Expense Load PVPBL > \$200,000

Expense Loading (4044 Appendix C):

The expense load is calculated as follows:

1. Number of valuation participants
2. PVPBL without load
3. Initial PBGC Annuity Interest Rate
4. $1\% + \{[(3) - 7.50\%] / 10\}$
5. Amount of load =

$$\$10,000 + [(2) - \$200,000] \times (4) + [\$200 \times (1)] =$$

6. **Loading percentage = (5) / (2)**

Small Plan DUEC Recovery Ratio - SPDRR

The average recovery ratio of PBGC's recovery experience for Due and Unpaid Employer Contributions (DUECs) over a five-year period.

$$\text{SPDRR} = \frac{\sum \text{DUEC Recoveries of all plans in SPDRR Period}}{\sum \text{DUEC of all plans in the SPDRR Period}}$$

Applies to plan with UNGB $\leq$ \$20 million and Termination Initiation Date (TID) of September 16, 2006 or later.

See [ATM Appendix H2](#) for SPDRR information

Calculation of Valuation Plan Assets using the SPDRR Example

- DOPT = 03/31/2013
- TID = 01/27/2012
- SPDRR = 30.72%
- Plan Assets as of DOPT: \$4,234,474.18
- Gross DUEC: \$612,044.00
- Net DUEC: \$ 612,044.00 x 30.72 % = \$188,019.92
- Valuation Assets: Plan Assets + Net DUEC
 - \$ 4,234,474.18 + \$ 188,019.92 = \$4,422,494.10

Objectives

- Assign Participants' Plan Benefits to the appropriate 6 Priority Categories
- Understand how assets affect benefits provided by PBGC
- Understand mechanics of asset allocation
- Know what to look for when deciding whether an asset allocation could affect your case, i.e., how to avoid actually conducting an asset allocation

Priority Classifications - Overview

- PC1 – Voluntary employee contributions
- PC2 – Mandatory employee contributions
- PC3 – Benefits in pay status or could have been in pay status 3 years prior to DOPT/BPD, based on the provisions that provide the lowest plan benefit within last 5 years prior to DOPT/BPD
- PC4 – Guaranteed benefits (with two exceptions)
- PC5 – Other vested (nonforfeitable) plan benefits
- PC6 – Other benefits under the plan

General Comments

- Priority Categories direct the order in which assets are allocated
 - They don't increase Plan benefits
 - They may increase PBGC-payable benefits
- The entire Plan benefit is allocated to at least one priority category.

General Comments

- ERISA §4044 asset allocations are done infrequently – only when the allocation could potentially produce a greater benefit than the guaranteed benefit.
- ERISA §4022(c) asset allocations are done in every case where at least one participant does not receive his/her full plan benefit.

General Comments

PBGC calculates four benefit amounts:

1. **Plan Benefit** – the amount provided under the plan's provisions
2. **Guaranteed Benefit** – The amount PBGC pays regardless of the amount of assets
3. **Title IV Benefit** – The guaranteed benefit plus the additional benefit provided by a §4044 allocation
4. **Termination Benefit** – The sum of the Title IV Benefit and the benefit provided by a §4022 (c) allocation

Priority Category One - PC1 Benefits

- Extremely rare
- Not guaranteed
- Not part of the Accrued Benefit
- Provided only through an asset allocation

Priority Category One - PC1 Benefits

- Voluntary Employee Contributions (EEC) with interest
- Plan allows participants to contribute a percentage of salary or some other amount
- Participant can elect to receive:
 - Lump Sum distribution = EEC + interest at DOTE/ARD
 - or
 - *Enhanced* annuity at ARD

Basic and NonBasic Type Benefits

Basic-Type Benefit (Guaranteeable) – a benefit that is guaranteed under § 4022 of ERISA or would be guaranteed if the limits in §§ 4022.22 - 4022.27 did not apply (CFR § 4001.2)

Nonbasic Type Benefit – any benefit provided by a plan other than a basic-type benefit (CFR §4001.2)

If assets fund a particular priority category, the assets are allocated to basic-type benefits first, then to nonbasic type benefits

Priority Category Two -

PC2 Benefits

- Mandatory Employee Contributions with interest
- Plan requires that the participant contribute in order to participate in the plan or receive an enhanced Employer provided benefit.
- Participant can elect to receive:
 - Lump Sum distribution = EEC + interest at DOPT
plus
reduced annuity at ARD
 - or
 - *Full* annuity at ARD

Priority Category Two - PC2 Benefits

To Calculate Benefits Due to Mandatory Employee Contribution use:

- Minimum Interest Rates (IR) specified by law credited to the Determination Date
 - Pre PY 1976 – no minimum (use plan rates)
 - PY 1976 to PY 1987 - 5%
 - PY 1988 & later - 120% of federal mid term rate
- Determination Date - the Earliest of (1) the date EECWI are withdrawn, (2) DOR, and (3) DOPT

Priority Category Two - PC2 Benefits

- Interest credited is projected from Determination Date to the Normal Retirement Date and converted to an annuity using:
 - 417(e)(3) rates in effect at the Determination Date
 - PY 2008 & later -- 3 segment rates and a mortality table projected to participant's retirement age

Priority Category Two - Valuing PC2 Benefits

Method depends on participants' election to withdraw or leave their MECs in the Plan.

If participant elects not to withdraw –

- PC2 value = PV of PC2 monthly benefit + PV of the PC2 pre-retirement death benefit

If participant elects to withdraw –

- PC2 value = Accumulated Mandatory Employee Contributions with Interest (AMEC) as of DOPT

The excess of AMEC over PV of PC2 monthly benefit + PV of PC2 pre-retirement death benefit is not guaranteed and is considered to be a nonbasic-type benefit

Priority Category Two

PC2 Benefits

- See [PBGC Operating Policy 5.11-1](#)
(Calculation of Priority Category 2 Benefits – Revised 08/30/2012)
- See [PBGC Operating Policy 5.11-2](#)
(Payment of Priority Category 2 Benefits – 03/28/2013)

Priority Category Three –

PC3 Benefits

Eligible Participants: Retired or who could have retired more than 3 years prior to DOPT/BPD

PC3 Benefit is calculated using:

- Plan provisions that provide the lowest benefit in the 5 year period prior to DOPT/BPD
- Participant's data as of earlier of DOTE and DOPT/BPD – 3
- Eligibility is determined as of DOPT/BPD-3 based on provisions in effect at DOPT/BPD-3
- Calculated based on earlier of ARD and first of the month following DOPT/BPD – 3

Priority Category Three – PC3 Benefits

- PBGC limits do not apply to PC3 benefits
 - Accrued at Normal Limit (for guaranteed purposes)
 - Maximum Guaranteeable Benefit
 - Phase-In
 - Majority Owner

Priority Category Three – PC3 Benefits

- Determine eligibility using **DOPT/BPD-3** plan provisions and service and age at DOPT/BPD-3
- Generally, calculate PC3 Benefits using service, compensation, age, as of DOPT/BPD-3, early retirement and form conversion factors at first of the month on or next following DOPT/BPD-3 using **DOPT/BPD-5** plan provisions since those are usually the provisions that provide the lowest benefit.
- If the participant's DOR is prior to DOPT/BPD-3, use DOR dates to calculate PC3 benefit.

Priority Category Three - PC3 Analysis

1. Determine whether the Participant is eligible for a PC3 benefit
2. Determine the amount of the PC3 Benefit
3. Determine the Present Value of the PC3 Benefit (include Load and QPSA, if any)
4. Sum the Present Value of all PC3 Benefits for the total plan PC3 liability

Priority Category Three - PC3 Analysis

5. Allocate assets to total PC3 liability
6. Determine the Plan-wide 4044 Funding Percent
7. Funded PC3 benefit for each participant
= Plan's 4044 Funding PC3 Percent x PC3 Benefit

Priority Category Three - PC3 Benefits

- PC3 benefit may be greater than guaranteed benefit if Plan benefit is reduced by:
 1. Accrued at Normal Limitation (ANL)
 2. Maximum Guaranteeable Benefit Limitation
 3. Majority Owner Phase-In Limitation

Priority Category Three - PC3 Benefits - Example 1

- Plan Information
 - DOPT: 11/18/2007
 - BPD: N/A
- Plan Formula:
 - Effective/Adopted 01/01/2000:
 - \$45 x Credited Service
 - Effective/Adopted 01/01/2005:
 - \$55 x Credited Service

Priority Category Three - PC3 Benefits - Example 1

- Plan Information
 - Credited Service: Years and Completed Months
 - Early Retirement: Age 55 with 5 years of Credited Service
 - Reduction For Early Retirement: 5% per year prior to Normal Retirement Date

Priority Category Three - PC3 Benefits - Example 1

- Participant A Information
 - Date of Birth: 11/09/1949
 - Date of Hire: 01/01/1995
 - Date of Termination: Active on DOPT
 - Normal Retirement Date: 12/01/2014

Priority Category Three - PC3 Benefits - Example 1

- Participant B Information
 - Date of Birth: 05/11/1945
 - Date of Hire: 11/01/1999
 - Date of Termination: Active on DOPT
 - Normal Retirement Date: 06/01/2010

Priority Category Three - PC3 Benefits - Example 1

- What is Participant A's and Participant B's Credited Service at DOPT-3 (11/18/2004)?
- What is Participant A's and Participant B's attained age at DOPT-3 (11/18/2004)?
- Is either Participant A or Participant B eligible for a PC3 Benefit (11/18/2004)?
- If eligible, what is Participant A's PC3 Benefit and Participant B's PC3 Benefit (12/01/2004)?

Priority Category Three - PC3 Benefits - Example 1

- Credited Service at DOPT-3 (11/18/2004)?
 - Participant A:
 - Participant B:
- Attained Age at DOPT-3 (11/18/2004)?
 - Participant A:
 - Participant B:
- Eligible for a PC3 Benefit (11/18/2004)?
 - Participant A:
 - Participant B:
- What is the PC3 Benefit (12/01/2004)?
 - Participant A:
 - Participant B:

Priority Category Three - PC3 Benefits – Answers Example 1

- Credited Service at DOPT-3 (11/18/2004)?
 - Participant A: **9.8333 Years**
 - Participant B: **5.0000 Years**
- Attained Age at DOPT-3 (11/18/2004)?
 - Participant A: **55**
 - Participant B: **59**
- Eligible for a PC3 Benefit (11/18/2004)?
 - Participant A: **Yes**
 - Participant B: **Yes**
- What is the PC3 Benefit (12/01/2004)?
 - Participant A: **\$221.25 = \$45 x 9.8333 x 0.5000**
 - Participant B: **\$163.13 = \$45 x 5.0000 x 0.7250**

Priority Category Three - PC3 Benefits - Example 2

DOPT: 10/18/2007

BPD: N/A

Assets as of DOPT: \$1,080,000

MGB at age 65, SLA = \$4,125.00

4 participants in plan

- PC3 eligibility determined as of 10/18/2004
- PC3 Benefit is calculated as if ARD = 11/01/2004 for ERFs and FCFs

Priority Category Three - PC3 Benefits - Example 2

DOPT: 10/18/2007

- AMB: Benefit Rate x YOBS

Benefit Rate	Effective/Adopted Date	Phase-In Limit
\$125	01/01/2002	100%
\$135	01/01/2004	\$60/60%
\$150	01/01/2006	\$20/20%

Priority Category Three - PC3 Benefits - Example 2

DOPT: 10/18/2007

- ERA: Age 55, 10 YOBS
- NRA: Age 65
- ERF: 5% per year
- Normal Form: Single Life Annuity
- Vesting: 5 Year Cliff

Priority Category Three - PC3 Benefits - Example 2

Participant 1:

- DOTE = 2/15/06; Sep Vested on DOPT
- Age 60 as of DOPT
- Years of Benefit Service (YOBS): 12 Years
- XRD = 11/01/2007
- $ERF = 1 - (.05 \times 5) = 0.75$
- Plan Benefit at XRD:
 - $\$150 \times 12 \times .75 = \$1,350$

Priority Category Three - PC3 Benefits - Example 2

Participant 1:

- Plan Benefit under 5 Year-old plan at XRD:
 - $\$125 \times 12 \times .75 = \$1,125$
- Plan Benefit under 3 Year-old plan at XRD:
 - $\$135 \times 12 \times .75 = \$1,215$
- Plan Benefit under 1 Year-old plan at XRD:
 - $\$150 \times 12 \times .75 = \$1,350$

Phase-In Limitation

Years in effect	Benefit Amount	Amount of Increase	Phase-in Limit	Phased-in Amount
5	\$1,125		100%	\$1,125
3	\$1,215	\$90	\$60/60%	\$60
1	\$1,350	\$135	\$20/20%	\$27
XRD - GMB				\$1,212

Priority Category Three - PC3 Benefits - Example 2

Participant 1:

1. Eligibility as of DOPT- 3 = 10/18/2004
 - Age 57
 - YOBS = 9 Years
 - Not enough service to be eligible for ER
 - No benefit in PC3

Priority Category Three - PC3 Benefits - Example 2

Participant 2:

- Active on DOPT
- Age 60
- Years of Benefit Service (YOBS): 16 Years
- XRD = 11/01/2007
- Plan Benefit: $\$150 \times 16 \times .75 = \$1,800$
- 5-Year old Plan Benefit at XRD = \$1,500
- 3-Year old Plan Benefit at XRD = \$1,620
- GMB at XRD = \$1,608

Priority Category Three - PC3 Benefits - Example 2

Participant 2:

1. Eligibility as of DOPT-3 = 10/18/2004
 - Age 57
 - YOBS = 13 Years
 - Eligible for ER
2. Determine the amount of the PC3 Benefit
 - AMB (under 5-year old plan) as of DOPT-3: $\$125 \times 13 = \$1,625$
 - Reduced by ERF: $(1 - (.05 \times \underline{8})) = .60$
 - PC3 Benefit: $\$1,625 \times .60 = \975

Priority Category Three - PC3 Benefits - Example 2

Participant 2:

3. Determine the PV of the PC3 Benefit

- Use the same PVF as the guaranteed benefit, i.e., PVF at age 60
- PV of PC3 Benefit:
 - $\$975 \times 12 \times 12.6294 = \$147,764$

Priority Category Three - PC3 Benefits - Example 2

Participant 3:

- Retired 8 years prior to DOPT
- Age 68 as of DOPT
- Benefit at ARD: \$2,625.00
- GMB = \$2,625.00

1. Eligibility as of DOPT-3 = 10/18/2004

- Met eligibility requirements because
DOR < DOPT - 3

Priority Category Three - PC3 Benefits - Example 2

Participant 3:

2. Determine the amount of the PC3 Benefit

- PC3 Benefit = Benefit in pay = \$2,625.00

3. Determine the PV of the PC3 Benefit

- Use the same PVF as the guaranteed benefit
- PV of PC3 Benefit:
 - $\$2,625.00 \times 12 \times 10.4263 = \$328,428$

Priority Category Three - PC3 Benefits - Example 2

Participant 4:

- DOR = 11/01/2006 at age 65
- Age 66 at DOPT
- Form of Annuity: SLA
- Years of Benefit Service (YOBS): 45 Years

Priority Category Three - PC3 Benefits - Example 2

Participant 4:

- Benefit at ARD: $\$150 \times 45 = \$6,750.00$
- 5-Year old Plan Benefit at ARD : $\$5,625.00$
- 3-Year old Plan Benefit at ARD : $\$6,075.00$
- MGB at DOPT (Age 65): $\$4,125.00$
- GMB = $\$4,125.00 \times 1.1000$ (Age 66)
= $\$4,537.50$

Priority Category Three - PC3 Benefits - Example 2

Participant 4:

1. Eligibility as of DOPT-3 = 10/18/2004
 - Age 63
 - YOBS = 43 Years
2. Determine the amount of the PC3 Benefit
 - AMB (under 5-year old plan) as of DOPT-3: $\$125 \times 43 = \$5,375$
 - Reduced by ERF: $(1 - (.05 \times \underline{2})) = .90$
 - PC3 Benefit: $\$5,375 \times .90 = \$4,837.50$

Priority Category Three - PC3 Benefits - Example 2

Participant 4:

3. Determine the PV of the PC3 Benefit

- Use the same PVF as the guaranteed benefit, i.e., age 66
- PV of PC3 Benefit:
 - $\$4,837.50 \times 12 \times 10.9936 = \$638,178$

Priority Category Three - PC3 Benefits - Example 2

Participant	Plan Benefit	GMB	PC3 Benefit	PV PC3
Separated Vested	\$1,350.00	\$1,212.00	\$0	\$0
Active	\$1,800.00	\$1,608.00	\$975.00	\$147,764
Retiree DOR = DOPT-8	\$2,625.00	\$2,625.00	\$2,625.00	\$328,428
Retiree DOR = DOPT-1	\$6,750.00	\$4,537.50	\$4,837.50	\$638,178
TOTAL				\$1,114,370

Priority Category Three - PC3 Benefits - Example 2

- Assets = \$1,080,000
- Gross PC3 = \$1,114,370 (assumed to include load and QPSA)
- Funding Percent =
 - $\$1,080,000 / \$1,114,370 = 96.9157\%$

Priority Category Three - PC3 Benefits - Example 2

	GMB	PC3 Benefit	Funded PC3 Benefit	Title IV Benefit = Greater of GMS and Funded PC3
1	\$1,212.00	\$0	\$0	\$1,212.00
2	\$1,608.00	\$975.00	\$944.93	\$1,608.00
3	\$2,625.00	\$2,625.00	\$2,544.04	\$2,625.00
4	\$4,537.50	\$4,837.50	\$4,688.30	\$4,688.30

Priority Category Four - PC4 Benefits

Guaranteed Benefits

Three things required as of DOPT/BPD

1. Vested (nonforfeitable) benefit
2. Pension benefit
3. Entitled

Priority Category Four - PC4 Benefits

- Gross PC4 Benefit = Guaranteed Benefit
- Net PC4 Benefit:
 - Guaranteed Benefit – PC3 Benefit

(Assume no benefits in PC1 and PC2)

Priority Category Four - PC4 Benefits

- **Accrued-at-Normal** – Plan benefit limited to the benefit amount payable at participant's NRA as a SLA (applies only to pre-NRA benefits)
- **Maximum Guaranteed Benefit** - Annuity amount adjusted at DOPT/BPD for early/late retirement and other forms of payment
- **Phase-in** larger of:
 - 20% for each full year the increase is in effect, or
 - \$20 for each full year the increase is in effect
 - not to exceed the actual increase

Priority Category Four - PC4 Benefits

Majority Owner Limitation:

- Owns 50% or more; 5-yr look back applies
- Guaranteed Benefit (GMB) =

GMB as if not a Majority Owner multiplied by a fraction =

Number years the plan has been in effect

10

If the later of the plan's original effective and adoption dates is 10 or more years prior to DOPT/BPD, the limitation will not affect Majority Owners' Guaranteed Benefit.

Priority Category Four - PC4 Benefits

- Majority Owner Limitation Allocation in PC4:

Two Tiers in PC4

- Tier 1 – Guaranteed Benefits plus Non-guaranteed benefits due to the Aggregate Limit
- Tier 2 – Non Guaranteed Benefits (Benefits lost due to Majority Owner Limitation)

Before Assets are allocated to Tier 2, all Guaranteed Benefits for all plan participants in Tier 1 must be fully funded.

Priority Category Four - PC4 Benefits – Example 1

DOPT – 10/18/2001

- AMB - \$50 x service on DOPT – 5; benefit rate increased \$10 each year, so DOPT Plan rate is \$100
- ERA – None
- NRA - Age 65
- Normal Single Form:
 - 10 Year Certain & Continuous (10C&C)

Priority Category Four - PC4 Benefits – Example 1

Benefit Rate	Effective/Adopted Date	Phase-in Limitation
\$50	10/01/1996	100%
\$60	10/01/1997	\$80/80%
\$70	10/01/1998	\$60/60%
\$80	10/01/1999	\$40/40%
\$90	10/01/2000	\$20/20%
\$100	10/01/2001	0%

Priority Category Four - PC4 Benefits – Example 1

- Participant A
 - Active on DOPT
 - Age 60
 - 35 years of service
- Monthly Vested Plan Benefit: $\$100 \times 35 = \$3,500$
- MIL - $\$3,392.05$ for DOPT= CY2001
 - $\text{MGB} = \$3,392.05 \times .9250(10\text{C}\&\text{C}) = \$3,137.65$
- Monthly vested accrued benefit - 5 year old plan
 $\$50 \times 35 = \$1,750$

Priority Category Four - PC4 Benefits – Example 1

4 year old plan

- $(\$60 \times 35) - (\$50 \times 35) = \$350 \leftarrow \text{Increase}$
- $\$350 \times 80\% = \$280 \leftarrow \text{Phased-in increase}$

3 year old plan

- $(\$70 \times 35) - (\$60 \times 35) = \$350 \leftarrow \text{Increase}$
- $\$350 \times 60\% = \$210 \leftarrow \text{Phased-in increase}$

Priority Category Four - PC4 Benefits – Example 1

2 year old plan

- $(\$80 \times 35) - (\$70 \times 35) = \$350 \leftarrow \text{Increase}$
- $\$350 \times 40\% = \$140 \leftarrow \text{Phased-in increase}$

Priority Category Four - PC4 Benefits – Example 1

1 year old plan

- $\$90 \times 35 = \$3,150$
- $\$3,392.05 \times .9250 = \$3,137.65 \leftarrow \text{MGB}$
- Limit Benefit to MGB
- $\$3,137.65 - (\$80 \times 35) = \$337.65 \leftarrow \text{Increase}$
- $\$337.65 \times 20\% = \67.53
- $\$67.53 \leftarrow \text{Phased-in increase}$

Priority Category Four - PC4 Benefits – Example 1

Gross PC4 Benefit

\$1,750.00	5 year old plan
\$280.00	4 year old plan
\$210.00	3 year old plan
\$140.00	2 year old plan
\$67.53	1 year old plan
<u>\$0.00</u>	< 1 year old plan
\$2,447.53	Guaranteed Benefit

Priority Category Five-

PC5 Benefits

- Other Vested Plan Benefits
 - Nonforfeitable Benefits limited in PC4 by:
 - Accrued-at-Normal Limit
 - Maximum Insurance Limit
 - Phase-in Limits
 - Vested, non-guaranteeable benefits
 - Post-DoPT automatic COLAs
 - Benefits accrued between BPD and DOPT

Priority Category Five-

PC5 Benefits

- Provided if plan assets sufficient beyond PC4
- PC5 is divided into subcategories:
 - Five year old plan first
 - then amendment by amendment
 - Ordered by the later of the effective date and the adoption date (Recognition Date). If more than one change has the same effective date, it is considered the same amendment and has equal priority in PC5.
- Use participant service and salary as of DOPT (not BPD) under each amendment

Priority Category Five- PC5 Benefits – Example

- Net PC5 benefit:
 - Gross PC5 – Gross PC4 =
 $\$3,500 - \$2,447.53 = \$1,052.47$
- \$1,052.47 is divided into Net PC5 Subcategories based on amendment dates
 - Gross PC5a = \$1,750 (5 year old provision)
 - Gross PC5b = \$2,100 (4 year old provision)
 - Gross PC5c = \$2,450 (3 year old provision)
 - Gross PC5d = \$2,800 (2 year old provision)
 - Gross PC5e = \$3,150 (1 year old provision)
 - Gross PC5f = \$3,500 (< 1 year old plan provision)

Priority Category Five- PC5 Benefits – Example

Net PC5a = Gross PC5a – Gross PC4 = \$1,750 - \$2,447.53 = \$0

Net PC5b = Gross PC5b – Gross PC5a = \$2,100 - \$2,447.53 * = \$0

Net PC5c = Gross PC5c – Gross PC5b = \$2,450 - \$2,447.53 = \$2.47

Net PC5d = Gross PC5d – Gross PC5c = \$2,800 - \$2,450 = \$350

Net PC5e = Gross PC5e – Gross PC5d = \$3,150 - \$2,800 = \$350**

Net PC5f = Gross PC5f – Gross PC5e = \$3,500 - \$3,150 = \$350

*When a Gross PC5 subcategory < Gross PC4, use Gross PC4

** When participants lose benefits due to both phase-in and MGB, allocate assets proportionally to that PC5 subcategory

Priority Category Five- PC5 Benefits – Example

Assets are allocated to subcategories by priority

- Net PC5a - 5 year old plan - \$0
 - Net PC5b - 4 year old plan - \$0
 - Net PC5c - 3 year old plan - \$2.47
 - Net PC5d - 2 year old plan - \$350.00
 - Net PC5e - 1 year old plan - \$350.00
 - Net PC5f - 0 year old plan - \$350.00
- \$1,052.47

Priority Category Six -

PC6 Benefits

All other benefits not in PC1 – PC5

- Non-vested accrued benefits
 - Active non-vested
 - Partially vested “active” participants (non-vested portion)
 - Vesting due to a Partial Termination (no amendment)
- Lump-Sum Death Benefit (pre/post retirement)
- Grow-in eligibilities
 - Active participants in ongoing employers only
 - Example – Unreduced benefit with 30 YOS

Priority Category Six - PC6 Benefits

- Available Assets are allocated to the total PC6 value regardless of the source of the values, i.e., no subcategories in PC6.

Assigning Plan Benefits to Priority Categories (PC) – Example 1

DOPT - 10/18/2001

- AMB : \$50 x YOS 5 Year old plan
 \$70 x YOS 2 Year old plan
 \$100 x YOS 1 Year old plan
- ERA: Age 55, 10 YOS; Any age with 30 YOS
- NRA: Age 65
- ERF: 5% per year
- Early Unreduced w/30 YOS
- Straight Life Annuity

Assigning Plan Benefits to the PC's Categories (PC) – Example 1

Maximum Guaranteeable Benefit

- Participant A
 - Active on DOPT
 - Age 60
 - 35 years of service
- Monthly Vested Benefit - 5 year old plan
 - $\$50 \times 35 \times 1.0000 \text{ (ERF)} = \$1,750$
- MGB at age 60 - \$1,500
- Guaranteed Benefit = \$1,500

Assigning Plan Benefits to the Priority Categories (PC)

– Example 1

PC3 Benefit

Step 1 - Is Participant A eligible for a PC3 Benefit?

- Age at DOPT- 3: 57
- Service as of DOPT - 3: 32 YOS

Step 2 - What is the PC3 monthly amount?

- Monthly vested accrued benefit at DOPT- 3 under 5 year old plan: $\$50 \times 32 = \$1,600$
- Monthly PC3 Benefit: \$1,600
(Eligible for Early Unreduced)

Assigning Plan Benefits to the Priority Categories (PC)

– Example 1

PC4 and PC5 Gross Benefit

Gross PC4 = \$1,500 (MGB)

Gross PC5a = \$1,750 = \$50 x 35 (5 year old plan)

Gross PC5b = \$2,450 = \$70 x 35 (2 year old plan)

Gross PC5c = \$3,500 = \$100 x 35 (1 year old plan)

Assigning Plan Benefits to the PCs

Net Monthly PCs – Example 1

PC6 = **\$0** = (no non-vested, death, or grow-in benefits)

Net PC5c = **\$1,050** = \$3,500 - \$2,450 (1 year old plan)

Net PC5b = **\$700** = \$2,450 - \$1,750 (2 year old plan)

Net PC5a = **\$150** = \$1,750 - \$1,600 (5 year old plan)

Net PC4 = **\$0** = \$1,500 - \$1,600

Net PC3 = **\$1,600**

PC2 = **\$0** (no contributions)

PC1 = **\$0** (no contributions)

Plan Benefit = \$3,500

Assigning Plan Benefits to the PCs – Example 1

Plan Assets = \$600,000

What is Participant A's Title IV Benefit?

Plan Liabilities (Net PCs): All Participants	
PC1 and PC2	\$0
Net PC3	\$1,000,000
Net PC4	\$2,000,000
Net PC5a	\$500,000
Net PC5b	\$ 750,000
Net PC5c	\$900,000
PC6	\$100,000

Assigning Plan Benefits to the PCs – Example 1

Plan Assets = \$600,000

What is Participant A's Title IV Benefit?

1. PC3 is 60% funded ($\$600,000 / \$1,000,000$)
2. Participant A's PC3 Benefit: \$1,600
3. Participant A's Funded PC3 Benefit: 60% of \$1,600 = \$960
4. Participant A's Guaranteed Benefit: \$1,500
5. Participant A's Title IV Benefit: \$1,500 = Greater of (3) and (4)

Assigning Plan Benefits to the PCs – Example 2

Plan Assets = \$1,600,000

What is Participant A's Title IV Benefit?

Plan Liabilities (Net PCs): All Participants	
PC1 and PC2	\$0
Net PC3	\$1,000,000
Net PC4	\$2,000,000
Net PC5a	\$500,000
Net PC5b	\$ 750,000
Net PC5c	\$900,000
PC6	\$100,000

Assigning Plan Benefits to the PCs – Example 2

Plan Assets = \$1,600,000

What is Participant A's Title IV Benefit?

1. PC3 is 100% funded ($\$1,000,000 / \$1,000,000$)
2. Net PC4 is 30% funded ($\$600,000^* / \$2,000,000$)
3. Participant A's PC3 Benefit: \$1,600
4. Participant A's Funded P3 Benefit: $\$1,600 = 1.0 \times \$1,600$
5. Participant A's Guaranteed Benefit: \$1,500
6. Participant A's Title IV Benefit: $\$1,600 = \text{Greater of (4) and (5)}$

*After allocating to PC3, \$600,000 ($\$1,600,000 - \$1,000,000$) remains to fund Net PC4



IN CLASS EXERCISES

Assigning Plan Benefits to the PCs

Example 3 – Exercise

- Participant A Information
 - PC3 Benefit = \$1,600
 - Guaranteed Benefit = \$1,500
 - Gross PC5a = \$1,750
 - Gross PC5b = \$2,450
 - Gross PC5c = \$3,500
 - No PC6 Liabilities

Asset Allocation

Example 3 – In Class Exercise

Plan Liabilities (Net PCs) : All Participants	
PC1 and PC2	\$0
Net PC3	\$1,000,000
Net PC4	\$2,000,000
Net PC5a	\$500,000
Net PC5b	\$750,000
Net PC5c	\$900,000
PC6	\$100,000

Asset Allocation

Example 3 – In Class Exercise

- What is Participant A's Title IV Benefit when Plan Assets:
 1. \$900,000?
 2. \$3,250,000?
 3. \$4,062,500?
 4. What is the minimum Plan Asset amount that would cause Participant A to receive a funded PC3 Benefit?

Answer to Exercise 3

Asset Allocation

Example 3 – In Class Exercise

What is Participant A's Title IV Benefit when Plan Assets are **\$900,000**?

- PC3 Funded benefit: $\$1,600 \times 90\% (\$900,000 / \$1,000,000) = \mathbf{\$1,440}$
- Guaranteed Benefit: **\$1,500**
- Title IV Benefit = Greater of \$1,440 and \$1,500 = **\$1,500**

Asset Allocation

Example 3 – In Class Exercise

What is Participant A's Title IV Benefit when Plan Assets are **\$3,250,000**?

- Assets Available for Net PC5a: **\$250,000** $\$3,250,000 - \$2,000,000 - \$1,000,000$
- Net PC5a Funded Benefit = **\$75**
 $\$150 \times 50\% (\$250,000 / \$500,000)$
- Title IV Benefit = **\$1,675** = $\$1,600 + \75

Asset Allocation

Example 3 – In Class Exercise

What is Participant A's Title IV Benefit when Plan Assets are **\$4,062,500**?

- Assets Available for Net PC5b: **\$562,500** $\$4,062,500 - \$500,000 - \$2,000,000 - \$1,000,000$
- Net PC5b Funded Benefit = **\$525**
 $\$700 \times 75\% (\$562,500 / \$750,000)$
- Title IV Benefit = **\$2,275** = $\$1,600 + \$150 + \$525$

Asset Allocation

Example 3 – In Class Exercise

- What is the minimum Plan Asset amount that would cause Participant A to receive a funded PC3 Benefit?
- Solve for y ; where $\$1,600 \times (y / \$1,000,000) > \$1,500$
- $y > \$937,503$



Example 4 – Plan Data

- DOPT – 10/18/2011
- Accrued Monthly Benefit
 - 5 Year old plan: 50% x AMC
 - 2 Year old plan: 60% x AMC
 - 1 Year old plan: 75% x AMC
- Form of Annuity: Straight Life Annuity
- Normal Retirement Age(NRA): Age 65
- Early Unreduced Retirement Age (EURA): Age 62 with 30 Years of Service



Example 4 – Participant Data

- AMC at DOPT = \$10,000
- AMC at DOPT-3 = \$9,500
- 100% Vested
- Age 65
- Active on DOPT
- Years of Service = 35



Example 4 – Guaranteed Monthly Benefit

- 5 Year old plan: $50\% \times \text{AMC} = \$5,000$
- 2 Year old plan: $60\% \times \text{AMC} = \$6,000$
- 1 Year old plan: $75\% \times \text{AMC} = \$7,500$
- 2011 MGB at age 65 as an SLA = \$4,500

- Guaranteed Monthly benefit = \$4,500



Example 4 – PC3 Benefits

- Age at DOPT-3: 62
- Service at DOPT-3: 32 years of service
- Eligible for PC3 Benefit?
- Monthly vested accrued benefit at DOPT-3
under 5 year old plan: $50\% \times \$9,500 = \$4,750$
- Monthly PC3 Benefit: \$4,750



Example 4 – PC4 and Gross PC5 Benefits

- Gross PC4 = \$4,500 (MGB)
- Gross PC5a = 50% x AMC = \$5,000 (5 year old plan)
- Gross PC5b = 60% x AMC = \$6,000 (2 year old plan)
- Gross PC5c = 75% x AMC = \$7,500 (1 year old plan)



Example 4 – Benefit in each Priority Category

Priority Category	Gross Benefit	Net Benefit	Present Value of all Net Benefits
1	\$0	\$0	\$0
2	\$0	\$0	\$0
3	\$4,750	\$4,750	\$300,000
4	\$4,500	\$0	\$0
5a	\$5,000	\$250	\$15,000
5b	\$6,000	\$1,000	\$60,000
5c	\$7,500	\$1,500	\$90,000
6	\$0	\$0	
	Total Plan Benefit	\$7,500	\$465,000



Example 4 – Funded Benefit in each PC

- Plan Assets fund 60% of PC3 Liability
- Participant's PC3 benefit = \$4,750
- Participant's Funded PC3 benefit = 60% of \$4,750 = \$2,850
- Participant's Guaranteed Monthly Benefit: \$4,500
- Title IV Benefit = \$4,500 because the guaranteed benefit > funded PC3 benefit



Example 4 – Greater Assets

- Assume only 1 participant in the plan and assets are \$313,275
- PC3 benefit fully funded = \$4,750
- Assets available to allocate to PC5a = $\$313,275 - \$300,000 = \$13,275$
- Percentage of funded PC5a = $\$13,275 / \$15,000 = 88.5\%$
- Funded PC5a benefit: $\$250 \times .885 = \221.25
- Title IV benefit = $\$4,750 + \$221.25 = \$4,971.25$

Avoiding Asset Allocations

Situations where Asset Allocations are not necessary:

- Plan has little or no Assets
- PBGC limits do not apply to any participant
- Plan Assets exhausted in PC4 and Guaranteed benefits exceed PC3 benefits for all participants
(Assume no Majority Owner or Aggregate Limit issues)

Questions??